

Evidence Room AP Agent OS — Professional

The AP Agent Operating System

Evidence Room AP Agent OS is the implementation-ready toolkit for building, governing, and scaling AI agents across Accounts Payable. It contains specifications for all 16 agents, complete governance frameworks, KPI definitions, testing methodologies, and business case models.

16-Agent Architecture

- **AP-INT-01 Invoice Intake Agent** — Autonomy L1: Review incoming invoices and determine whether information is complete, correctly extracted and suitable for downstream processing.
- **AP-VAL-02 Invoice Validation Agent** — Autonomy L2: Validate invoice header and line data against business rules, master data and compliance requirements.
- **AP-MAT-03 Matching Agent** — Autonomy L2: Perform PO, price, quantity and receipt matching with tolerance analysis for multi-line invoices.
- **AP-TRI-04 Exception Triage Agent** — Autonomy L2: Classify every AP exception into the defined taxonomy and recommend the next action.
- **AP-GR-05 Goods Receipt Agent** — Autonomy L2: Identify missing receipts, determine likely responsible employees and prepare follow-ups.
- **AP-POQ-06 PO Quality Agent** — Autonomy L1: Identify poor PO creation practices driving downstream AP exceptions.
- **AP-APR-07 Approval Agent** — Autonomy L2: Monitor approval workflows and identify stalls, delegation issues and deadline risks.
- **AP-SUP-08 Supplier Resolution Agent** — Autonomy L2: Draft supplier communications for invoice discrepancies, missing information and statement differences.

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- **AP-INT-09 Internal Follow-Up Agent** — Autonomy L2: Draft and manage internal follow-ups for missing GR, incorrect PO, approvals, coding and requester clarification.
- **AP-DUP-10 Duplicate & Anomaly Agent** — Autonomy L1: Identify exact and near duplicates, suspicious patterns and supplier anomalies. Not a fraud-detection guarantee.
- **AP-VSR-11 Vendor Statement Reconciliation Agent** — Autonomy L2: Reconcile supplier statements against AP records; identify missing items and timing differences.
- **AP-PPR-12 Payment Proposal Review Agent** — Autonomy L1: Perform pre-payment analytical review. Payment authorisation remains human-controlled.
- **AP-CLS-13 AP Close Agent** — Autonomy L1: Support month-end AP activities including completeness review, accruals and cut-off.
- **AP-RPT-14 AP Reporting Agent** — Autonomy L3: Produce daily, weekly and monthly AP operational and performance reporting.

- **AP-RCA-15 Root Cause Agent** — Autonomy L1: Analyse recurring exceptions and determine systemic causes across suppliers, PO discipline, systems and master data.
- **AP-ORC-16 AP Manager / Orchestrator** — Autonomy L2: Supervisory layer across the agent workforce — task distribution, performance monitoring, escalation and management reporting.

5-Stage Responsibility Model

- Level 0 — Observe: Agent reviews but cannot act. All outputs are logged for analysis.
- Level 1 — Recommend: Agent produces recommendations for humans. No system actions.
- Level 2 — Prepare: Agent prepares actions (drafts, classifications, routing) requiring human approval.
- Level 3 — Execute within guardrails: Agent executes pre-approved low-risk activities within defined tolerances.
- Level 4 — Managed autonomy: Agent operates independently within boundaries with exception-based human oversight.

Governance Framework

Every agent requires: job description, inputs, tools, responsibilities, exclusions, human owner, approval requirements, escalation criteria, output standard, control requirements, audit evidence, KPIs, performance history, autonomy level, failure handling, and cost monitoring.

Exception Taxonomy

27 exception categories defined with root cause, resolution path, responsible party, assigned agent, automation potential, and risk level. See accompanying Exception Tracker spreadsheet.

Implementation Roadmap

10-phase deployment framework from baseline through scale. Accelerated timeline: 4-6 weeks for one well-bounded agent. Actual duration depends on systems, controls, integrations, data quality, and governance maturity.

Evidence Standard

Industry benchmarks sourced from Ardent Partners, Forrester, PwC, and Deloitte. See Research Ledger for full citations. Evidence Room frameworks are independently authored and clearly distinguished from third-party data.